SOCIOL 723: Social Statistics II

Week 1 (Tuesday), The Linear Model and OLS

Wenhao Jiang
Department of Sociology, Fall 2026

Duke
UNIVERSITY

Today

Orientation

OLS in Scalar Form

Statistical Properties of OLS

What Is OLS Estimating?

Regression as Adjustment

Tuesday: everything in scalar form, through *Regression as Adjustment*. **Thursday**: the matrix form, from zero, plus lab.

Orientation

Three Pillars of this Course

- **Estimation** (Weeks 1–4). What is the estimator, and what are its statistical properties? OLS, then maximum likelihood as the general engine.
- **Causal inference** (Weeks 5–11). What quantity are we after, and what would have to be true for our estimate to recover it?
- **Machine learning** (Weeks 12–13). What do we do when the number of covariates is large and the functional form is unknown?

The through-line

Nearly everything in this course is some version of one idea: *compare units that are alike in all the ways that matter*. Regression is the first and most transparent implementation of that idea.

How to Read These Slides

The two-star convention, used all semester

- A frame title marked ★ is **essential**: you must be able to explain and use this material. Exam questions concentrate here; study these frames first.
- A frame title marked ★ is here *for your understanding*, not for evaluation. Derivations in matrix form, projection geometry, and asymptotic arguments all fall in this category.
- Unmarked frames are ordinary lecture material: fair game on problem sets and exams as *supporting* material, at a lighter level than the ★ frames.

- **What I will examine**: the scalar versions of everything, the algebra you can do by hand with $\sum_i$ signs, plus interpretation, assumptions, and knowing which tool fits which problem.
- My slides are deliberately dense so that they work as a reference after class. Stop me whenever something is unclear; that is what lecture is for.

What You Already Know: Model Comparison

In *Social Statistics I* you compared a **compact** model to an **augmented** model:

$$\text{MODEL C:} \quad Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$$
$$\text{MODEL A:} \quad Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

and asked whether the extra parameters bought enough reduction in error:

$$PRE = \frac{SSE_C - SSE_A}{SSE_C}, \qquad F = \frac{(SSE_C - SSE_A)/(PA - PC)}{SSE_A/(n - PA)}.$$

Refresher: SSE = sum of squared errors; PRE = *proportional reduction in error*; PA, PC = number of parameters in models A and C. F is the usual F statistic: under H_0 (extra parameters useless), $F \sim F_{PA-PC,\, n-PA}$, and "enough" means F beats its critical value.

What You Already Know: Model Comparison

In *Social Statistics I* you compared a **compact** model to an **augmented** model:

$$\begin{aligned} \text{MODEL C:} \quad & Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i \\ \text{MODEL A:} \quad & Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i \end{aligned}$$

and asked whether the extra parameters bought enough reduction in error:

$$PRE = \frac{SSE_C - SSE_A}{SSE_C}, \qquad F = \frac{(SSE_C - SSE_A)/(PA - PC)}{SSE_A/(n - PA)}.$$

Refresher: SSE = sum of squared errors; PRE = *proportional reduction in error*; PA, PC = number of parameters in models A and C. F is the usual F statistic: under H_0 (extra parameters useless), $F \sim F_{PA-PC,\, n-PA}$, and "enough" means F beats its critical value.

- All of this stays. Today we look *inside* the estimator: where do $\hat{\beta}_0, \hat{\beta}_1$ come from, and what do they mean?
- Thursday rewrites it in matrix form: we mostly work in scalars, but the compact notation is what modern papers and methods use.

Notation We Will Use All Semester

Population versus sample

$$E[\,\cdot\,] : \text{the } \textbf{population} \text{ expectation, a fixed, unknown number}$$

$$\mathbb{E}_n[\,\cdot\,] \equiv \frac{1}{n}\sum_{i=1}^{n}(\,\cdot\,) : \text{the } \textbf{sample} \text{ (empirical) expectation, a random variable}$$

- Greek letters (β, σ, ϵ) are *population* quantities we never observe; hats ($\hat{\beta}$, $\hat{e}$) are *sample* quantities we compute.
- ϵ_i always denotes the error term; $\hat{e}_i$ the fitted residual. They are different objects, and confusing them is the single most common source of trouble.
- Later: bold italic = vector or matrix (lowercase $\boldsymbol{y}$ for vectors, uppercase $\boldsymbol{X}$ for matrices), $'$ = transpose.

The whole of statistical inference is the study of the gap between E and $\mathbb{E}_n$.

Scalar OLS

The Bivariate Model

The Bivariate Linear Model

For each unit $i = 1, \ldots, n$ we observe an outcome Y_i and one regressor X_i; suppose the population relationship satisfies the linear model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad E[\epsilon_i \mid X_i] = 0,$$

so the average of Y at each value of X is the line itself: $E[Y_i \mid X_i = x] = \beta_0 + \beta_1 x$.

- β_0, the **intercept**: the population average of Y when $X = 0$, since $E[Y \mid X = 0] = \beta_0$.
- β_1, the **slope**: the difference in average Y between units one unit apart in X.
- ϵ_i, the **error**: everything else that moves unit i's outcome off its group average; mean zero at every X by assumption.

Running example, all semester

GSS 2010–2022, full-time workers aged 25–64 ($n = 3{,}509$).
Y_i = log personal earnings, X_i = years of schooling completed.

The Least Squares Criterion ★

We do not observe β_0, β_1. We pick estimates b_0, b_1 to make the fitted line as close as possible to the data. "Close" = small squared vertical distances:

$$S(b_0, b_1) \;=\; \sum_{i=1}^{n} \big(Y_i - b_0 - b_1 X_i\big)^2.$$

The Least Squares Criterion ★

We do not observe β_0, β_1. We pick estimates b_0, b_1 to make the fitted line as close as possible to the data. "Close" = small squared vertical distances:

$$S(b_0, b_1) \;=\; \sum_{i=1}^{n} \big(Y_i - b_0 - b_1 X_i\big)^2.$$

- Why squares? They punish large misses more than small ones, they make the problem differentiable, and, as we will see, they deliver the conditional *mean*.
- Why vertical? Because we are predicting Y from X, not the reverse. Regressing X on Y gives a different line.

The OLS estimates are $(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{b_0, b_1} S(b_0, b_1)$.

Step 1: The First-Order Conditions

S is a smooth, convex function of two numbers. Set both partial derivatives to zero. Using the chain rule,

$$\frac{\partial S}{\partial b_0} = \sum_{i=1}^{n} 2(Y_i - b_0 - b_1 X_i) \cdot (-1) \overset{!}{=} 0,$$

$$\frac{\partial S}{\partial b_1} = \sum_{i=1}^{n} 2(Y_i - b_0 - b_1 X_i) \cdot (-X_i) \overset{!}{=} 0.$$

Dividing by -2 gives the two normal equations:

$$\sum_i (Y_i - \hat\beta_0 - \hat\beta_1 X_i) = 0, \tag{N1}$$

$$\sum_i X_i (Y_i - \hat\beta_0 - \hat\beta_1 X_i) = 0. \tag{N2}$$

What the Normal Equations Say ★

Write $\hat{e}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ for the fitted residual. Then

$$(\text{N1}): \quad \sum_i \hat{e}_i = 0 \qquad\qquad (\text{N2}): \quad \sum_i X_i \hat{e}_i = 0.$$

- The residuals **sum to zero**. Hence $\bar{\hat{e}} = 0$, and the fitted line passes through the point of means $(\bar{X}, \bar{Y})$.
- The residuals are **uncorrelated with the regressor**: given (N1), (N2) is equivalent to $\sum_i (X_i - \bar{X})\hat{e}_i = 0$.

Read this carefully

These are *mechanical consequences of the algebra*, true in every sample, for every dataset. Plotting $\hat{e}_i$ against X_i and finding no correlation is therefore not evidence that the model is correct.

Uncorrelated, Mean-Independent, Independent ★

Three statements about an error and a regressor, strictly ordered:

Statement	In words	What it rules out
$\mathrm{Cov}(X, \epsilon) = 0$	no *linear* pattern	a tilted cloud, nothing more
$E[\epsilon \mid X] = 0$	no pattern in the *average*	any curve in the mean
$\epsilon \perp\!\!\!\perp X$	no pattern in the *distribution*	heteroskedasticity too

- **Uncorrelated but not mean-independent**: fit a line to a quadratic relationship. The correlation with X is zero, yet the errors' *average* traces the leftover curve: positive at the extremes, negative in the middle. A residual plot catches this; the correlation cannot.
- **Mean-independent but not independent**: $E[\epsilon \mid X] = 0$ at every x, but the *spread* of ϵ changes with X: heteroskedasticity, Week 2's subject.
- Our model assumes the middle rung. The normal equations deliver only the bottom one, and for *sample* residuals, mechanically.

Step 2: Solving for $\hat{\beta}_0$

Divide (N1) by n:

$$\frac{1}{n}\sum_i Y_i - \hat{\beta}_0 - \hat{\beta}_1 \frac{1}{n}\sum_i X_i = 0 \quad \Longrightarrow \quad \bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X},$$

so

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}}$$

- The intercept is whatever it must be for the line to pass through $(\bar{X}, \bar{Y})$.
- Immediate consequence: if you *center* the regressor ($X_i^c = X_i - \bar{X}$), the intercept becomes $\bar{Y}$. Centering never changes the slope, only what the intercept means.

Step 3: Solving for $\hat{\beta}_1$ ★

Substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ into (N2):

$$\sum_i X_i \left(Y_i - \bar{Y} + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i \right) = 0$$

$$\sum_i X_i (Y_i - \bar{Y}) = \hat{\beta}_1 \sum_i X_i (X_i - \bar{X}).$$

Step 3: Solving for $\hat{\beta}_1$ ★

Substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ into (N2):

$$\sum_i X_i \Big(Y_i - \bar{Y} + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i \Big) = 0$$

$$\sum_i X_i (Y_i - \bar{Y}) = \hat{\beta}_1 \sum_i X_i (X_i - \bar{X}).$$

Now use $\sum_i \bar{X}(Y_i - \bar{Y}) = 0$ and $\sum_i \bar{X}(X_i - \bar{X}) = 0$, which let us replace X_i by $(X_i - \bar{X})$ on both sides:

$$\boxed{\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\mathrm{Cov}_n(X, Y)}{\mathrm{Var}_n(X)}}$$

Reading the Slope Formula ★

$$\hat{\beta}_1 = \frac{\mathrm{Cov}_n(X,Y)}{\mathrm{Var}_n(X)} = \mathrm{Corr}_n(X,Y) \cdot \frac{s_Y}{s_X} = \sum_i w_i \, \frac{Y_i - \bar{Y}}{X_i - \bar{X}}, \quad w_i = \frac{(X_i - \bar{X})^2}{\sum_j (X_j - \bar{X})^2}.$$

- **Numerator**: do X and Y move together across units?
- **Denominator**: how much does X vary? No variation in X, no slope; the formula is $0/0$.
- The third form is worth remembering: the OLS slope is a *weighted average of unit-by-unit slopes*, with more weight on observations whose X is far from $\bar{X}$. Extreme X values matter disproportionately.

The Running Example, by Hand

GSS 2010–2022, $n = 3{,}509$, Y = log earnings, X = years of schooling:

$$\bar{X} = 14.48, \quad \bar{Y} = 9.950, \quad \mathrm{Cov}_n(X, Y) = 0.9476, \quad \mathrm{Var}_n(X) = 7.981.$$

The Running Example, by Hand

GSS 2010–2022, $n = 3{,}509$, Y = log earnings, X = years of schooling:

$$\bar{X} = 14.48, \quad \bar{Y} = 9.950, \quad \mathrm{Cov}_n(X, Y) = 0.9476, \quad \mathrm{Var}_n(X) = 7.981.$$

$$\hat{\beta}_1 = \frac{0.9476}{7.981} = 0.1187,$$
$$\hat{\beta}_0 = 9.950 - 0.1187 \times 14.48 = 8.231.$$

Interpretation

Comparing two full-time workers who differ by one year of schooling, the one with more schooling has about 11.9% higher earnings *on average*.

Note the words: *comparing*, *on average*. Not "one more year **causes** 11.9% higher earnings."

Interpretation: Units, Logs, and Nonlinearity

- Y in levels, X in levels: β_1 is a change in Y-units per X-unit.
- $\log Y$**,** X **in levels** (our case): $\beta_1 \approx$ the *proportional* change in Y. Exactly, $100(e^{\beta_1} - 1)\%$; for $\beta_1 = 0.1187$ this is 12.6%. The approximation is good below about 0.2.
- $\log Y$, $\log X$: β_1 is an elasticity.
- **Quadratics**: with $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$, the marginal effect is $\beta_1 + 2\beta_2 X$, it depends on where you evaluate it. Report it at meaningful values, not only at the mean.
- **Interactions**: with $\beta_3 X_i Z_i$, the effect of X is $\beta_1 + \beta_3 Z$. The "main effect" β_1 is the effect at $Z = 0$, which may be outside the data.

Multiple Regression, in Scalar Form

Adding a Second Regressor

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i.$$

Minimizing $S(b_0, b_1, b_2) = \sum_i (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i})^2$ gives *three* normal equations:

$$\sum_i \hat{e}_i = 0, \qquad \sum_i X_{1i}\hat{e}_i = 0, \qquad \sum_i X_{2i}\hat{e}_i = 0.$$

- One equation per parameter. With k parameters you get k equations in k unknowns.
- Solving three equations by substitution is unpleasant but possible; with ten it is hopeless. That bookkeeping problem, nothing deeper, is what Thursday's matrix notation tidies up. Everything we *use* this semester stays scalar.

What "Holding Constant" Actually Means ★

$\hat{\beta}_1$ is usually described as "the effect of X_1 *holding* X_2 *constant*." Where does that come from?

Partialling out: a three-step recipe

1. Regress X_{1i} on X_{2i} (and a constant). Keep the residuals $\tilde{X}_{1i} = X_{1i} - \hat{\gamma}_0 - \hat{\gamma}_1 X_{2i}$.
2. Regress Y_i on X_{2i}. Keep the residuals $\tilde{Y}_i$.
3. Regress $\tilde{Y}_i$ on $\tilde{X}_{1i}$. The slope you get is exactly $\hat{\beta}_1$ from the multiple regression.

This is the **Frisch–Waugh–Lovell** theorem. In scalar form:

$$\hat{\beta}_1 = \frac{\sum_i \tilde{X}_{1i} \tilde{Y}_i}{\sum_i \tilde{X}_{1i}^2} = \frac{\sum_i \tilde{X}_{1i} Y_i}{\sum_i \tilde{X}_{1i}^2}.$$

Why the Recipe Works, Slowly: Raw Y First

Claim: regressing *raw* Y_i on the residualized $\tilde{X}_{1i}$ already recovers the multiple-regression slope. Substitute the fitted long regression $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{e}_i$ into the numerator:

$$\sum_i \tilde{X}_{1i} Y_i = \hat{\beta}_0 \underbrace{\sum_i \tilde{X}_{1i}}_{=0} + \hat{\beta}_1 \sum_i \tilde{X}_{1i} X_{1i} + \hat{\beta}_2 \underbrace{\sum_i \tilde{X}_{1i} X_{2i}}_{=0} + \underbrace{\sum_i \tilde{X}_{1i} \hat{e}_i}_{=0} .$$

- First and third zeros: $\tilde{X}_1$ is a fitted residual from the regression of X_1 on X_2, so it sums to zero and is orthogonal to its own regressor X_2: the normal equations, again.
- Last zero: $\hat{e}$ is orthogonal to 1, X_1, and X_2, and $\tilde{X}_1 = X_1 - \hat{\gamma}_0 - \hat{\gamma}_1 X_2$ is built from exactly those three.
- Middle term: $\sum_i \tilde{X}_{1i} X_{1i} = \sum_i \tilde{X}_{1i} \big(\tilde{X}_{1i} + \hat{\gamma}_0 + \hat{\gamma}_1 X_{2i} \big) = \sum_i \tilde{X}_{1i}^2$.

Divide through: $\sum_i \tilde{X}_{1i} Y_i \big/ \sum_i \tilde{X}_{1i}^2 = \hat{\beta}_1$. ∎

Residualizing Y Too Changes Nothing

Now replace Y_i by $\tilde{Y}_i = Y_i - \hat{\delta}_0 - \hat{\delta}_1 X_{2i}$, the residual from regressing Y on X_2:

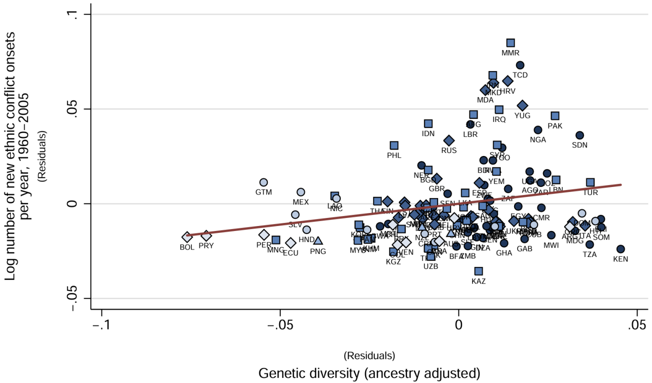
$$\sum_i \tilde{X}_{1i}\tilde{Y}_i = \sum_i \tilde{X}_{1i} Y_i - \hat{\delta}_0 \underbrace{\sum_i \tilde{X}_{1i}}_{=0} - \hat{\delta}_1 \underbrace{\sum_i \tilde{X}_{1i} X_{2i}}_{=0} = \sum_i \tilde{X}_{1i} Y_i.$$

The part removed from Y is a function of X_2 alone, and $\tilde{X}_1$ is orthogonal to anything built from X_2. Same numerator, same denominator, same slope:

$$\hat{\beta}_1 = \frac{\sum_i \tilde{X}_{1i}\,\tilde{Y}_i}{\sum_i \tilde{X}_{1i}^2} = \frac{\sum_i \tilde{X}_{1i}\,Y_i}{\sum_i \tilde{X}_{1i}^2}.$$

- Why residualize Y at all, then? For the *picture*: plotting $\tilde{Y}$ against $\tilde{X}_1$ shows the relationship with X_2 removed from both axes. Next frame.
- What you cannot change is the denominator: it must be $\sum_i \tilde{X}_{1i}^2$. Regressing $\tilde{Y}$ on *raw* X_1 divides by too large a number and attenuates the slope. (Problem Set 1.)

FWL in Practice: the Added-Variable Plot



Ethnic conflict onsets against genetic diversity, *both axes residualized* on the paper's full control set (Arbatli, Ashraf, Galor, and Klemp 2020).

- Both axes say "(Residuals)": this is $\tilde{Y}$ against $\tilde{X}_1$, and by FWL the slope of the fitted line *is* the paper's multivariate coefficient.
- The *added-variable plot*: how applied papers draw a many-control regression in two dimensions, and the only honest way to do it.
- In R: `car::avPlots()`, or residualize by hand as in lab.

Why This Matters

- "Controlling for X_2" means throwing away every scrap of X_1 that X_2 can predict, and using only what is left.
- If X_1 and X_2 are strongly related, very little is left. The coefficient is then estimated from a thin, possibly unrepresentative slice of the data, and its standard error will say so.
- In our data $sd(\texttt{educ}) = 2.82$; after residualizing on experience it is 2.73. Little is lost, so the two coefficients on `educ` are similar (0.119 vs. 0.138).

Omitted Variables

Short and Long Regressions ★

Suppose the "long" regression is

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \epsilon_i,$$

but we run the "short" one, omitting W_i:

$$Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_i + u_i.$$

Let δ_1 be the slope from the *auxiliary* regression of the omitted variable on the included one: $W_i = \delta_0 + \delta_1 X_i + \nu_i$.

Short and Long Regressions ★

Suppose the "long" regression is

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \epsilon_i,$$

but we run the "short" one, omitting W_i:

$$Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_i + u_i.$$

Let δ_1 be the slope from the *auxiliary* regression of the omitted variable on the included one: $W_i = \delta_0 + \delta_1 X_i + \nu_i$.

Omitted variable bias

$$\boxed{\tilde{\beta}_1 \;=\; \beta_1 \;+\; \beta_2\,\delta_1}$$

"short = long + (effect of omitted) × (regression of omitted on included)"

Derivation, in Scalar Form

Apply the bivariate slope formula to the short regression, then substitute the long model for Y_i. Everything below is *sample* projection algebra: the long regression's residual is orthogonal to its own regressors by construction, which is why the last term is exactly zero.

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\mathrm{Cov}_n(X,Y)}{\mathrm{Var}_n(X)} = \frac{\mathrm{Cov}_n\left(X,\ \beta_0 + \beta_1 X + \beta_2 W + \epsilon\right)}{\mathrm{Var}_n(X)} \\
&= \beta_1 \underbrace{\frac{\mathrm{Cov}_n(X,X)}{\mathrm{Var}_n(X)}}_{=1} + \beta_2 \underbrace{\frac{\mathrm{Cov}_n(X,W)}{\mathrm{Var}_n(X)}}_{=\delta_1} + \underbrace{\frac{\mathrm{Cov}_n(X,\epsilon)}{\mathrm{Var}_n(X)}}_{=0 \text{ in the long regression}} \\
&= \beta_1 + \beta_2\, \delta_1. \qquad \blacksquare
\end{aligned}$$

Constants drop out of covariances, which is why β_0 never appears.

Signing the Bias ★

	$\mathrm{Cov}_n(X, W) > 0$	$\mathrm{Cov}_n(X, W) < 0$
$\beta_2 > 0$	positive bias (*overstate*)	negative bias (*understate*)
$\beta_2 < 0$	negative bias (*understate*)	positive bias (*overstate*)

- You need *both* channels. An omitted variable uncorrelated with X costs precision, not consistency.
- Omit ability from a schooling–earnings regression: ability raises earnings ($\beta_2 > 0$) and rises with schooling ($\delta_1 > 0$) $\Rightarrow$ the return to schooling is overstated.
- With several omitted variables you cannot generally sign the total bias from theory alone; the terms can offset.

Omitted Variable Bias in the GSS

Log earnings on schooling; full-time workers 25–64, $n = 3{,}509$.

	(1)	(2)	(3)
Years of schooling	0.119	0.140	0.118
Experience, experience^2, female, Black		✓	✓
Vocabulary score (WORDSUM), parental education			✓
R^2	0.129	0.233	0.245

Check the formula on column (3) → (2). The two omitted variables contribute

$$\underbrace{0.0406}_{\hat{\beta}_{\text{wordsum}}} \times \underbrace{0.281}_{\hat{\delta}_{\text{wordsum}}} + \underbrace{0.0205}_{\hat{\beta}_{\text{pareduc}}} \times \underbrace{0.527}_{\hat{\delta}_{\text{pareduc}}} = 0.0222 = 0.1405 - 0.1183. \quad \checkmark$$

But how much is *still* missing? The algebra is silent. That question is what Weeks 5–11 are about.

Fit

Decomposing the Variation ★

Write $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ so that $Y_i = \hat{Y}_i + \hat{e}_i$. Subtract $\bar{Y}$, square, and sum:

$$\underbrace{\sum_i (Y_i - \bar{Y})^2}_{TSS = SSE_C} = \underbrace{\sum_i (\hat{Y}_i - \bar{Y})^2}_{ESS} + \underbrace{\sum_i \hat{e}_i^2}_{RSS = SSE_A} + 2\underbrace{\sum_i (\hat{Y}_i - \bar{Y})\hat{e}_i}_{=0}.$$

The cross term vanishes because $\sum_i \hat{e}_i = 0$ and $\sum_i X_i \hat{e}_i = 0$, and $\hat{Y}_i$ is a linear function of X_i.

$TSS/ESS/RSS$ = *total/explained/residual* sum of squares. Careful: other books use SSR for "regression" *or* for "residual"; when in doubt, say the words, not the letters.

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \qquad \Longleftrightarrow \qquad PRE = \frac{SSE_C - SSE_A}{SSE_C}.$$

R^2 is exactly the PRE of your model against the intercept-only model. In our example $TSS = 3062.3$, $RSS = 2667.6$, so $R^2 = 0.129$.

A Warning About R^2

- R^2 never decreases when you add a regressor. Adjusted R^2 penalizes k, but is still not a principled model-selection rule.
- For causal work R^2 is nearly irrelevant: a randomized experiment with $R^2 = 0.01$ identifies the effect; an observational regression with $R^2 = 0.9$ identifies nothing in particular.

Use it for

Prediction quality, and comparing nested specifications on the *same* sample. Not for adjudicating causal claims.

Checkpoint: What You Should Be Able to Do

1. Write down the least-squares objective and differentiate it to get the normal equations.
2. State what the normal equations imply about the residuals, and explain why those implications are not evidence of anything.
3. Derive and interpret $\hat{\beta}_1 = \mathrm{Cov}_n(X, Y)/\,\mathrm{Var}_n(X)$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$.
4. Describe partialling out in three steps and say what “holding constant” costs.
5. Derive $\tilde{\beta}_1 = \beta_1 + \beta_2\delta_1$ and sign the bias.
6. Connect R^2 to PRE.

We can now compute $\hat{\beta}_1$ from any dataset. Next question: before we trust it, how does it behave across samples, and what do the assumptions buy us?

Properties

Before the List: What Are We After?

We can compute $\hat{\beta}_1$; the question is *how good* it is. $\hat{\beta}_1$ came from one sample, and a different sample would have produced a different number. So we ask how it behaves *across* samples:

- **Accuracy**: is $\hat{\beta}_1$ centred on the target, or systematically off? (Unbiasedness.)
- **Precision**: how much does it move from sample to sample? (Variance, and what drives it.)
- **Inference**: can we quantify how far off we are likely to be? (The sampling distribution, the heart of Week 2.)

The role of the assumptions

None of these guarantees is free. The assumptions on the next frame are the price list: each buys a specific property. Knowing which buys what tells you exactly what breaks when an assumption fails, and the rest of the course largely lives in those failures.

The Classical Assumptions ★

A1 Linearity in parameters: $Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_{k-1} X_{k-1,i} + \epsilon_i$. (Not restrictive: X may contain logs, powers, splines, interactions.)

A2 Random sampling: (Y_i, X_i) i.i.d. from the population.

A3 No perfect collinearity: X varies, and no regressor is an exact linear combination of the others (Thursday: $\text{rank}(\boldsymbol{X}) = k$).

A4 Strict exogeneity: $E[\epsilon_i \mid X_i] = 0$.

A5 Spherical errors: $\text{Var}(\epsilon_i \mid X) = \sigma^2$ (homoskedasticity) and $\text{Cov}(\epsilon_i, \epsilon_j \mid X) = 0$ for $i \neq j$ (no correlation across units).

A6 Normality (optional): $\epsilon_i \mid X_i \sim \mathcal{N}(0, \sigma^2)$.

A1, A4–A6: *population*. A2: sampling. A3: your *sample*, the only one checkable in data.

What each buys you

A1–A4 $\Rightarrow$ unbiasedness; +A5 $\Rightarrow$ the variance formula and Gauss–Markov; +A6 $\Rightarrow$ exact t and F. Week 2 drops A5; Weeks 5+ interrogate A4, the one that matters for causality.

Unbiasedness, in Scalar Form ★

Take the bivariate case. Start from the slope formula and substitute $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\sum_i (X_i - \bar{X})\big(\beta_1 (X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon})\big)}{\sum_i (X_i - \bar{X})^2}$$

$$= \beta_1 + \frac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}.$$

Unbiasedness, in Scalar Form ★

Take the bivariate case. Start from the slope formula and substitute $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\sum_i (X_i - \bar{X})\big(\beta_1 (X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon})\big)}{\sum_i (X_i - \bar{X})^2}$$

$$= \beta_1 + \frac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}.$$

Remember this line

$\hat{\beta}_1 = \beta_1 +$ (a weighted average of the errors). Every result about bias and variance in this course comes from staring at it.

No assumption on $\bar{\epsilon}$: it multiplies $\sum_i (X_i - \bar{X}) = 0$, so it vanishes by algebra. Taking expectations conditional on X and using A4 ($E[\epsilon_i \mid X] = 0$) gives $E[\hat{\beta}_1 \mid X] = \beta_1$, and hence $E[\hat{\beta}_1] = \beta_1$.

The Variance of $\hat{\beta}_1$, in Scalar Form ★

From $\hat{\beta}_1 - \beta_1 = \dfrac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}$, and treating X as fixed:

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sum_i (X_i - \bar{X})^2 \ \mathrm{Var}(\epsilon_i \mid X)}{\left[\sum_i (X_i - \bar{X})^2\right]^2}$$

using only that the ϵ_i are uncorrelated across units.

The Variance of $\hat{\beta}_1$, in Scalar Form ★

From $\hat{\beta}_1 - \beta_1 = \dfrac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}$, and treating X as fixed:

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sum_i (X_i - \bar{X})^2 \ \mathrm{Var}(\epsilon_i \mid X)}{\left[\sum_i (X_i - \bar{X})^2\right]^2}$$

using only that the ϵ_i are uncorrelated across units.

Under homoskedasticity (A5), $\mathrm{Var}(\epsilon_i \mid X) = \sigma^2$ factors out:

$$\boxed{\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}}$$

Keep the first expression in view. Next week we simply refuse to assume $\mathrm{Var}(\epsilon_i \mid X)$ is constant, and that is all a “robust standard error” is.

What Drives Precision ★

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} = \frac{\sigma^2}{n\ \mathrm{Var}_n(X)},$$

and with more regressors, $\mathrm{Var}(\hat{\beta}_j) = \dfrac{\sigma^2}{n\ \mathrm{Var}_n(X_j)\,(1 - R_j^2)}$ where R_j^2 is from regressing X_j on the other regressors.

- **Noise** σ^2 up $\Rightarrow$ variance up.
- **Sample size** n up $\Rightarrow$ variance down like $1/n$; standard error down like $1/\sqrt{n}$.
- **Spread of** X up $\Rightarrow$ variance down.
- **Collinearity** $R_j^2 \to 1 \Rightarrow$ variance explodes; the *variance inflation factor* is $1/(1 - R_j^2)$.

Collinearity violates no assumption. It says the data contain little independent information about β_j.

Estimating σ^2

$\sigma^2 = E[\epsilon_i^2]$ is unknown; we have residuals, not errors. The natural estimator is

$$s^2 = \frac{1}{n-k}\sum_i \hat{e}_i^2, \qquad E[s^2] = \sigma^2.$$

- Why $n-k$ and not n? The residuals satisfy k linear constraints (the normal equations), so they are systematically "too short."
- Thursday's matrix language proves this in one line ★; for now, take $E[\sum_i \hat{e}_i^2] = \sigma^2(n-k)$ on trust and note that it is exact, not approximate.
- Then $\hat{V}(\hat{\beta}_1) = s^2 / \sum_i (X_i - \bar{X})^2$, whose square root is the standard error your software prints.

Reading Gauss–Markov Correctly

The theorem, in words

Under A1–A5 (hence their name: the *Gauss–Markov assumptions*; A6 is not needed), among all *linear unbiased* estimators OLS has the smallest variance: it is the **B**est **L**inear **U**nbiased **E**stimator (BLUE). Thursday states and proves this in matrix form.

- "Best" only within that class. Biased estimators (ridge, lasso, shrinkage) can have far lower mean squared error, Week 12.
- Efficiency requires A5. Under heteroskedasticity OLS stays unbiased but is no longer BLUE; weighted least squares with the correct weights is.
- In practice we rarely know the weights, so we keep OLS and fix the *standard errors* instead, the entire subject of next week.
- Gauss–Markov says nothing about causality. It is a statement about the variance of an estimator for the population regression coefficient, whatever that coefficient happens to mean.

Finite-Sample Distribution Under Normality

If A6 also holds, then conditional on X,

$$\hat{\beta}_1 \sim \mathcal{N}\Big(\beta_1,\ \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}\Big), \qquad \frac{(n-k)s^2}{\sigma^2} \sim \chi^2_{n-k}, \qquad \hat{\beta}_1 \perp\!\!\!\perp s^2,$$

so that $t = \dfrac{\hat{\beta}_1 - \beta_1}{\mathrm{se}(\hat{\beta}_1)} \sim t_{n-k}$.

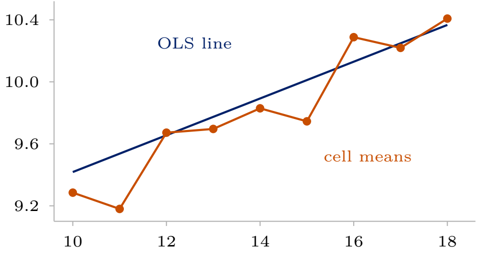
- $\hat{\beta}_1$ is a *linear* function of the errors, hence normal when they are.
- Without A6, all of this holds only *approximately*, and only for large n. That is next week.

Every guarantee in this section leaned on the assumption that the average of Y at each X lies on a line in the population. Next we drop it, and ask what $\hat{\beta}_1$ estimates when no line is true.

CEF and BLP

Nobody Believes the Line

Regress `lnearn` on `educ` in our GSS extract: $\hat{\beta}_1 = 0.119$. Now plot the *actual* mean of `lnearn` at each value of `educ`.



step	actual	the line says
$11 \to 12$	$+0.49$	$+0.12$
$12 \to 13$	$+0.02$	$+0.12$
$15 \to 16$	$+0.54$	$+0.12$
$16 \to 17$	-0.07	$+0.12$

The returns are concentrated at the diploma years, 12 and 16. The line spreads them evenly across all years. It misses the shape entirely: it has no idea that 12 and 16 are special.

And yet we reported 0.119 anyway. What is that number?

Estimating *What*, Exactly?

The circularity

The least-squares mechanics were pure sample algebra, true for any data. But everything in the last section, unbiasedness, the variance formula, Gauss–Markov, leaned on one population assumption: that the average of Y at each X lies on a line. That assumption is what defined β_1, and it is what delivered $E[\hat{\beta}_1] = \beta_1$. If it is false, and the previous frame showed that it is, there is no true β_1 for $\hat{\beta}_1$ to be unbiased *for*.

- **Estimand**: the population quantity you want. **Estimator**: the recipe. **Estimate**: the number.
- We have the last two. We have never named the first without assuming it into existence.
- Unbiasedness, consistency, standard errors are all statements about distance from a target. No target, no statements.

The Plan

Define β as a feature of the joint distribution of (Y_i, X_i) , something that exists in every population, whether or not any line is true.

1. The **CEF** $E[Y_i \mid X_i]$: what we would most like to know. Assumption-free, and the answer to most sociological questions as actually posed.
2. The **best linear predictor**: the line closest to the CEF. Defined for any joint distribution, no linearity assumed, none needed.
3. A theorem connecting them, which tells us exactly what 0.119 is a summary *of*.

Why this is worth an hour

It converts "your model is misspecified" from a fatal objection into a statement about *approximation error*, which you can then evaluate, bound, or fix by saturating. You cannot defend a regression you cannot say the target of.

The Conditional Expectation Function (CEF)

Everything so far was about a *sample*. Step back to the population.

For an outcome Y_i and covariates X_i, the conditional expectation function is

$$g(x) \;=\; E[Y_i \mid X_i = x].$$

- In words: the average outcome among all units in the population with $X_i = x$, average earnings among people with exactly 16 years of schooling.
- This is what most sociological questions are really about: *how does the average of Y move as we move through the population along X?*
- Because X_i is random, $E[Y_i \mid X_i]$ is itself a random variable.

Three Facts About the CEF ★

1. Law of iterated expectations

$E[Y_i] = E\big[E[Y_i \mid X_i]\big]$, the overall mean is the average of the group means.

Three Facts About the CEF ★

1. Law of iterated expectations

$E[Y_i] = E\big[E[Y_i \mid X_i]\big]$, the overall mean is the average of the group means.

2. CEF decomposition

Define $\epsilon_i \equiv Y_i - E[Y_i \mid X_i]$. Then $Y_i = E[Y_i \mid X_i] + \epsilon_i$ with $E[\epsilon_i \mid X_i] = 0$, and hence $E[\epsilon_i m(X_i)] = 0$ for *any* function $m(\cdot)$.

Three Facts About the CEF ★

1. Law of iterated expectations

$E[Y_i] = E\big[E[Y_i \mid X_i]\big]$, the overall mean is the average of the group means.

2. CEF decomposition

Define $\epsilon_i \equiv Y_i - E[Y_i \mid X_i]$. Then $Y_i = E[Y_i \mid X_i] + \epsilon_i$ with $E[\epsilon_i \mid X_i] = 0$, and hence $E[\epsilon_i m(X_i)] = 0$ for *any* function $m(\cdot)$.

3. CEF prediction property

$E[Y_i \mid X_i] = \arg\min_{m(\cdot)} E\big[(Y_i - m(X_i))^2\big]$: among *all* functions of X_i, the CEF is the best mean-squared-error predictor.

These are identities, not assumptions. They carry no causal content whatever.

Proof of the Decomposition Property ★

By definition $\epsilon_i = Y_i - E[Y_i \mid X_i]$, so

$$E[\epsilon_i \mid X_i] = E\big[Y_i \mid X_i\big] - E[Y_i \mid X_i] = 0.$$

For any function $m(\cdot)$, apply the law of iterated expectations:

$$E\big[\epsilon_i\, m(X_i)\big] = E\Big[\,E\big[\epsilon_i\, m(X_i) \mid X_i\big]\Big] = E\Big[m(X_i)\ \underbrace{E[\epsilon_i \mid X_i]}_{=0}\Big] = 0. \quad \blacksquare$$

Any random variable splits into a part *explained* by X_i and a part orthogonal to *every* function of X_i.

Proof of the Prediction Property ★

For any candidate predictor $m(X_i)$, add and subtract the CEF:

$$
\begin{aligned}
E\left[(Y_i - m(X_i))^2\right] &= E\Big[\{(Y_i - E[Y_i \mid X_i]) + (E[Y_i \mid X_i] - m(X_i))\}^2\Big] \\
&= \underbrace{E\left[(Y_i - E[Y_i \mid X_i])^2\right]}_{\text{does not involve } m} + \underbrace{E\left[(E[Y_i \mid X_i] - m(X_i))^2\right]}_{\geq 0} \\
&\quad + 2\ \underbrace{E\left[\epsilon_i \left(E[Y_i \mid X_i] - m(X_i)\right)\right]}_{= 0 \text{ by decomposition}}.
\end{aligned}
$$

The middle term is minimized at zero by setting $m(X_i) = E[Y_i \mid X_i]$. ■

The Population Regression, or Best Linear Predictor

The CEF is a general function and typically unknown. Restrict attention to *linear* predictors and define

$$\boldsymbol{\beta} = \arg\min_{b} E\big[(Y_i - X_i'b)^2\big] \quad \Longrightarrow \quad E\big[X_i(Y_i - X_i'\boldsymbol{\beta})\big] = 0.$$

This is the population counterpart of the normal equations. In the bivariate case it is exactly what you expect:

$$\beta_1 = \frac{\mathrm{Cov}(X_i, Y_i)}{\mathrm{Var}(X_i)}, \qquad \beta_0 = E[Y_i] - \beta_1 E[X_i].$$

Defining $e_i \equiv Y_i - X_i'\boldsymbol{\beta}$, we get $E[X_i e_i] = 0$ by construction, in every population. Again: algebra, not causality.

Reading note for Tuesday: read $X_i'\boldsymbol{b}$ simply as $b_0 + b_1 X_i$; Thursday introduces the general k-regressor version.

Why Should We Care About the Best *Linear* Predictor? ★

Three standard justifications (Angrist and Pischke 2009):

1. **If the CEF is linear**, the population regression *is* the CEF.
2. **If the CEF is nonlinear**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear approximation to it: $\boldsymbol{\beta} = \arg\min_b E\left[(E[Y_i \mid X_i] - X_i'b)^2\right]$.
3. **Regardless**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear predictor of Y_i.

Why Should We Care About the Best *Linear* Predictor? ★

Three standard justifications (Angrist and Pischke 2009):

1. **If the CEF is linear**, the population regression *is* the CEF.
2. **If the CEF is nonlinear**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear approximation to it: $\boldsymbol{\beta} = \arg\min_b E\big[(E[Y_i \mid X_i] - X_i'b)^2\big]$.
3. **Regardless**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear predictor of Y_i.

Justification 2 is the useful one in practice. OLS is a *well-defined summary of the CEF* even when you do not believe the CEF is a line, so "the model is wrong" is not, by itself, a fatal objection.

Proof of the Regression-CEF Theorem ★

Claim: $\boldsymbol{\beta} = \arg\min_b E[(Y_i - X_i'b)^2]$ also solves $\arg\min_b E\left[(E[Y_i \mid X_i] - X_i'b)^2\right]$.

$$
\begin{aligned}
E[(Y_i - X_i'b)^2] &= E\Big[\big\{(Y_i - E[Y_i \mid X_i]) + (E[Y_i \mid X_i] - X_i'b)\big\}^2\Big] \\
&= \underbrace{E\left[(Y_i - E[Y_i \mid X_i])^2\right]}_{\text{free of } b} + E\left[(E[Y_i \mid X_i] - X_i'b)^2\right] \\
&\quad + 2\, \underbrace{E\left[\epsilon_i \left(E[Y_i \mid X_i] - X_i'b\right)\right]}_{=0}
\end{aligned}
$$

The two objectives differ by a constant, so they share a minimizer. ■

Saturated Models: When Regression Is the CEF

Suppose X_i is *discrete* with J cells. Include a full set of $J-1$ dummies plus a constant. The regression is then saturated: one free parameter per cell.

- A saturated regression reproduces the CEF *exactly*, cell by cell. No functional-form assumption is being made at all.
- Example: regress log earnings on a dummy for each value of `educ`. The fitted values *are* the cell means.
- Contrast: entering `educ` linearly imposes a constant increment, and is therefore an approximation.

Practical upshot

Whenever you can afford the degrees of freedom, saturating in the key covariate is a free robustness check on functional form. We do this in lab.

What That Detour Bought Us: Two Questions, Kept Apart ★

Every argument about a regression coefficient is one of two questions. Keep them apart and most confusion dissolves.

Q1. What does $\hat{\beta}_1$ estimate?

The minimum-MSE linear approximation to the CEF.

Always answerable, algebra, true in every population.

"Your model is misspecified" attacks this, and usually *fails*: OLS has an honest answer.

Q2. Is *that* the causal effect?

Only if the units being compared are comparable.

Never answerable by algebra, a claim about how the data came to be.

"Your treated and controls differ" attacks this, and usually *lands*: nothing can repair it.

What That Detour Bought Us: Two Questions, Kept Apart ★

Every argument about a regression coefficient is one of two questions. Keep them apart and most confusion dissolves.

Q1. What does $\hat{\beta}_1$ estimate?

The minimum-MSE linear approximation to the CEF.

Always answerable, algebra, true in every population.

"Your model is misspecified" attacks this, and usually *fails*: OLS has an honest answer.

Q2. Is *that* the causal effect?

Only if the units being compared are comparable.

Never answerable by algebra, a claim about how the data came to be.

"Your treated and controls differ" attacks this, and usually *lands*: nothing can repair it.

Weeks 1–4 are Q1. Weeks 5–13 are Q2. That is the whole course.

Adjustment

Regression With a Binary Treatment

Let $D_i \in \{0, 1\}$ be a treatment and W_i controls:

$$Y_i = \alpha + \rho D_i + W_i' \gamma + e_i.$$

(Read $W_i'\gamma$ as: each control enters with its own coefficient.) Suppose that *within cells of* W_i treatment is as good as randomly assigned. Then the cell-specific difference in means

$$\tau(w) = E[Y_i \mid D_i = 1, W_i = w] - E[Y_i \mid D_i = 0, W_i = w]$$

is the causal effect in that cell.

Question: when $\tau(w)$ varies across cells, what single number does OLS report?

OLS Reports a Variance-Weighted Average ★

If the model is saturated in W_i (Angrist and Pischke 2009):

$$\rho = \frac{E\left[\sigma_D^2(W_i)\,\tau(W_i)\right]}{E\left[\sigma_D^2(W_i)\right]}, \qquad \sigma_D^2(w) = \mathrm{Var}(D_i \mid W_i = w) = p(w)\big(1 - p(w)\big),$$

where $p(w) = \Pr(D_i = 1 \mid W_i = w)$ is the *propensity score.*

- OLS averages cell-level effects with weights proportional to the conditional variance of treatment.
- Cells with $p(w) \approx 0.5$ dominate; cells where nearly everyone is (or is not) treated get almost no weight.
- So OLS does not generally deliver the ATE or the ATT, it delivers its own weighted average. Matching and weighting (Week 6) let you choose the weights deliberately.

Good Controls, Bad Controls ★

Adding a control is not automatically an improvement.

Good controls

- Confounders: common causes of D and Y.
- Pre-treatment covariates that predict Y (precision gains).

Bad controls

- Post-treatment variables (mediators): conditioning removes part of the effect you want.
- Colliders: common effects of D and Y; conditioning *creates* association where none existed.
- Proxies for the outcome.

Preview

No algebra distinguishes these cases: the distinction is *causal*, not statistical. We need a language for causal assumptions, which is Week 5.

Where Tuesday Leaves Us

- Everything today used only scalar algebra: the least-squares recipe, its statistical properties under the classical assumptions, what it estimates when no line is true (the CEF and its best linear approximation), and what regression can and cannot say about causality.
- **Thursday** we say the same things again in matrix form: no new statistics, only a new notation. We will mostly keep working in scalars, but the compact notation is what you will meet in modern papers and methods. We start from zero; no linear algebra is assumed.

Before Thursday

Install `R` and `RStudio`; skim MHE Chapter 3 and CCI Chapter 6.

References

- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 3]
- Morgan, Stephen L. and Christopher Winship. 2015. *Counterfactuals and Causal Inference*, 2nd ed. Cambridge University Press. [Ch. 6]
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning*, 2nd ed. Springer. [Ch. 2–3]
- Arbatlı, Cemal Eren, Quamrul H. Ashraf, Oded Galor, and Marc Klemp. 2020. "Diversity and Conflict." *Econometrica* 88(2): 727–797. [the added-variable plot example]
- Correll, Josh, Abigail M. Folberg, Charles M. Judd, Gary H. McClelland, and Carey S. Ryan. 2024. *Data Analysis: A Model Comparison Approach to Regression, ANOVA, and Beyond*, 4th ed. Routledge. [for review]